

Heterogeneity in Macro

Course Syllabus – M2 Quantitative Economics

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Format	Three sessions, nine contact hours in total; lectures and hands-on code laboratories.
Language	English.
Period	November 2026 (provisional); exact dates and session durations to be confirmed.
Materials	Three handouts, executable model packages and guided exercises.
Course page	https://vermandel.org/gea-heterogeneity/

Course description

When do differences across households change the answer to a macroeconomic question? This course moves from a transparent asset-market equilibrium to three quantitative frameworks: a life-cycle overlapping-generations economy, an Aiyagari economy with uninsured earnings risk, and a one-asset Heterogeneous-Agent New Keynesian (HANK) economy. The common organizing principle is the feedback between individual choices, the distribution of states, aggregation and equilibrium prices.

The course emphasizes model selection and controlled experiments rather than complexity for its own sake. Students learn to distinguish a household consumption response from an equilibrium policy effect, explain why numerical equilibrium requires more than individual Euler equations, and connect economic equations to Dynare code. The HANK session contrasts monetary and fiscal transmission: similar aggregate monetary responses can conceal different household mechanisms, while fiscal timing and incidence can change consumption demand substantially. This is a conditional comparison, not a general claim that heterogeneity is irrelevant for monetary policy.

Learning outcomes

By the end of the course, students should be able to:

1. Explain when aggregation is valid and when a distribution is needed to determine prices and policy responses.
2. Derive household budgets, Euler conditions and borrowing constraints; close an economy with production, government accounting and market clearing.
3. Distinguish life-cycle saving from precautionary saving and identify the asset that households hold in each model.
4. Explain the three components of a HA stationary calculation: individual policies, an invariant distribution, and an equilibrium or calibration loop.
5. Map equations into Dynare, run an existing model, modify a controlled parameter or policy experiment, and report results with correct timing and units.
6. Interpret policy functions, wealth distributions, MPCs and impulse responses, and check feasibility, accounting and numerical robustness.

Prerequisites and preparation

Students should be familiar with intermediate macroeconomics (the consumption Euler equation, production and capital accumulation), constrained optimization, expectations and elementary Markov chains. Basic MATLAB skills—running scripts, editing parameters, loading results and plotting—are expected; prior experience writing a HA solver is not. RBC and New Keynesian concepts are reviewed where they enter the models.

Before the first laboratory, download the code packages, install the software below and check that MATLAB can find Dynare. Work in a copy of the supplied model folder and retain an unchanged reference run. Each experiment should state a prediction, the exact change, what is held fixed, and an economic interpretation.

Session 1 – Market clearing and life-cycle heterogeneity

Question: how can individual saving decisions and equilibrium prices be jointly determined when households differ?

- A selective map of heterogeneity; representative-agent aggregation, two-household asset demand, and zero-net-supply bond clearing.
- From finite households to distributions and productive capital: the household–prices feedback.
- An 80-cohort OLG economy with age-dependent earnings, retirement, consumption and saving; stationary equilibrium and annual calibration.
- Dynare’s macro-processor: declarations, loops, conditional statements and generated equations.
- Nonlinear perfect-foresight productivity and balanced age-targeted redistribution experiments; a no-borrowing variant and fixed-price household MPCs.

Practice: trace an Euler equation and an aggregator into the code, inspect the macro expansion, modify a supplied experiment, and distinguish a transfer response from a household MPC. The handout includes six code exercises. Richer demography and portfolio structures are discussed, not implemented by a one-line parameter change.

Session 2 – Uninsured income risk, Aiyagari and HA-RBC

Question: how do incomplete insurance and precautionary saving generate a wealth distribution and affect equilibrium capital and prices?

- Historical motivation: Huggett, Aiyagari and Krusell–Smith; computational bottlenecks and more efficient approximations.
- Infinitely lived households, two-state earnings risk, one asset, a borrowing constraint, competitive firms and capital-market clearing.
- Policies and the stationary distribution; solve the equilibrium rate at fixed preferences versus calibrate patience to a target return.
- Dynare heterogeneous and aggregate blocks; time iteration, complementarity, endogenous-grid intuition and forward distribution iteration.
- Consumption/saving policies, the wealth distribution, MPCs and a small aggregate productivity shock; grid and horizon checks.

Practice: identify blocks, compute invariant income shares, verify budgets and resource balance, interpret a windfall, and compare numerical grids or shock persistence. Six guided exercises are provided. The appendix develops a simplified Krusell–Smith-style aggregate-shock extension and sequence-space intuition. Aiyagari already contains capital; the extension adds aggregate fluctuations. This calibration uses illustrative model periods, not the quarterly units of Session 3.

Session 3 – HANK in practice: monetary and fiscal policy

Question: when does household heterogeneity change policy transmission, and through which channels?

- A selective review of HANK, microeconomic MPC evidence, monetary redistribution and intertemporal fiscal transmission.
- A quarterly one-liquid-asset economy: uninsured earnings risk, common hours, competitive goods production, sticky nominal wages, government bonds, a Taylor rule and fiscal financing. There is no productive capital in this benchmark.
- Theoretical blocks alongside the actual Dynare listings; stationary calibration, loading a consistent steady state and first-order aggregate dynamics.
- Quarterly and annual windfall MPCs; monetary impulse responses; a matched RANK comparison and direct/income-channel decomposition.
- Two fiscal extensions: immediate versus debt financing, and uniform versus earnings-targeted transfers. Additional sensitivity to the Taylor rule.

Practice: exercises 1–4 form the core lab (block mapping, monetary timing, MPC horizons and financing); exercises 5–7 cover targeting, the Taylor rule and decomposition. Students save each scenario under a new label. Normann’s sticky-price/individual-labor model is contrasted in an appendix; two-asset HANK is a further direction, not a validated additional classroom model.

Software and reproducibility

- **MATLAB and Dynare 7.x are required. Use version 7.1 or later within the 7.x series; 7.2 is recommended.** Sessions 2–3 were validated with MATLAB R2025a and Dynare 7.2. The original Session 1 OLG validation used 7.0, but a single 7.2 installation is recommended for the full course.
- **Do not use Dynare 7.0 for Sessions 2–3.** Its heterogeneous-block complementarity conditions have a sign bug fixed in 7.1; see the [official release notes](#).
- Python 3 and the packages in each `requirements.txt` are used for the Session 2–3 figures and independent checks. MATLAB/Dynare solve the equilibrium; Python does not supply a substitute model.
- To rebuild the handouts rather than simply read the PDFs, LaTeX, latexmk and the minted/Pygments tooling are needed. These are not prerequisites for manipulating the model code.

The handouts share a notation reference. Assets and capital are dated at the end of the period; distributions describe inherited states. Model-specific code aliases and different time units are documented explicitly. Numerical convergence must be accompanied by budget, market-clearing and distribution checks. Exercises need not rerun every robustness calculation during class.

Exercises and assessment

The nine contact hours combine exposition and selected guided tasks; not every extension is intended to be completed in class. Remaining exercises provide consolidation and self-study. A useful lab submission consists of a short prediction, a reproducible code change, a labelled figure or numerical check, and an interpretation distinguishing partial from general equilibrium.

The formal assessment format, grading weights and submission deadlines have not yet been specified. The suggested lab outputs above describe learning activities, not an announced grading scheme. Instructor corrections for Sessions 2–3 are retained separately from the public student downloads.

Reading guide

The supplied handouts are the main reading. The following are selective background or follow-up references; students are not expected to read every paper before the course.

- **Life cycle and aggregation:** Samuelson (1958), *An Exact Consumption-Loan Model of Interest*; Diamond (1965), *National Debt in a Neoclassical Growth Model*.
- **Incomplete markets:** Huggett (1993), *The Risk-Free Rate in Heterogeneous-Agent Incomplete-Insurance Economies*; Aiyagari (1994), *Uninsured Idiosyncratic Risk and Aggregate Saving*; Krusell and Smith (1998), *Income and Wealth Heterogeneity in the Macroeconomy*.
- **Numerical methods:** Carroll (2006), *The Method of Endogenous Gridpoints*; Reiter (2009), *Solving Heterogeneous-Agent Models by Projection and Perturbation*; Auclert, Bardóczy, Rognlie and Straub (2021), *Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models*.
- **HANK and policy:** McKay, Nakamura and Steinsson (2016), *The Power of Forward Guidance Revisited*; Kaplan, Moll and Violante (2018), *Monetary Policy According to HANK*; Auclert (2019), *Monetary Policy and the Redistribution Channel*; Auclert, Rognlie and Straub (2024), *The Intertemporal Keynesian Cross*.
- **Pedagogical sources:** Valerio Pieroni's *Topics in Macroeconomics* notes (HA and HANK chapters) and Normann Rion's *Heterogeneity in Dynare 7* masterclass (2026). The course supplies adapted teaching models, not unmodified replications of either source.

Full bibliographic details and the additional recent literature are provided in each handout.